

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
STATEMENT OF WORK (SOW)

Instruction of virtual API 520/521 Pressure Relieving Systems training course

- Background and objectives – This SOW is to define the requirements involved in procuring virtual training covering specific sections of API 520 (Sizing, Selection, and Installation of Pressure-relieving Devices) and API 521 (Pressure-relieving and Depressuring Systems)
- Description of work to be performed – NASA procurement encompasses training specific to API 520/521 Pressure Relieving Systems training course. Total instruction time is approximately 24 hours (6 days, 4 hours per day).
- Schedule – To be determined between June - August 2025
- Contractor responsibilities:

Training on API 520 and 521 to include the following:

- Introduction and Basic Terminology
- Pressure Relief Valve (PRV) Design and Operation
- PRD Codes and Standards
- PRV Sizing and Selection: Vapor, Liquid, and Steam
- Two-Phase PRV Sizing Methods
- Installation Requirements, including 3% Rule
- Relief System Design Philosophy
- Use of Isolation Valves
- Overpressure Scenarios and Determination of Relief Loads
 - Fire and Blocked Discharge
 - Control Valve Failure
 - Thermal Relief and Heat Exchanger Tube Rupture
 - Distillation Columns and Utility Failures
 - Chemical Reactions and Explosions
- Rupture Disks: Usage and Installation
- Overpressure Protection of Atmospheric: Storage Tank
- Overpressure Protection of Power Boilers (ASME Section I)

Requester

4/22/2025

Date

Institutional Safety Training and Excellence
Title

NASA Safety Center
Office